

Class 09: Candy Mini Project

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Background

There will be more analysis in PCA in this document

The following will be just a loading of the candy data.

Data import

This data is originally gathered from the 538 website.

```
candy_file <- "https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-ran
# save the data link into a variable candy_file

candy = read.csv(candy_file, row.names=1)
#use read.csv to save the data into a data frame variable 'candy', and eliminates the first l

head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

```
#test the results of candy to ensure proper data frame was made
```

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

85 candy types total.

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

38 total fruity candy types are in this dataset.

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["Reese's Peanut Butter cup", ]$winpercent
```

```
[1] 84.18029
```

Reese's Peanut Butter cups have a winpercent of 84.18029.

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat",]$winpercent
```

```
[1] 76.7686
```

76.7686 is the winpercent value for kitkat.

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

49.6535 is the value for tootsie roll snack bars.

```
# install.packages("skimr")
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Only winpercent seems to have a different scale to the other columns in the candy dataset.

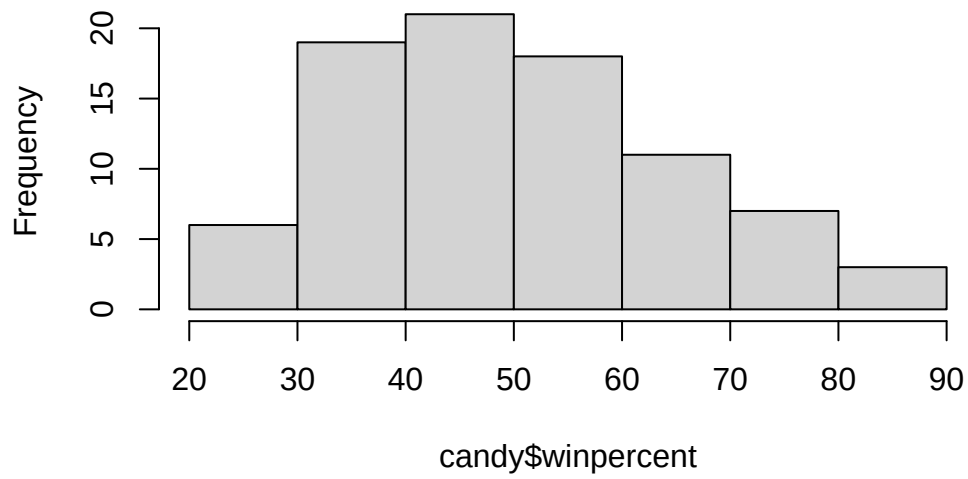
Q7. What do you think a zero and one represent for the candy\$chocolate column?

They represent whether there are any gaps in the dataset for the specific variable, for example if the chocolate had an empty space for rating for a person it would have a slightly lower complete_rate and slightly higher n_missing.

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

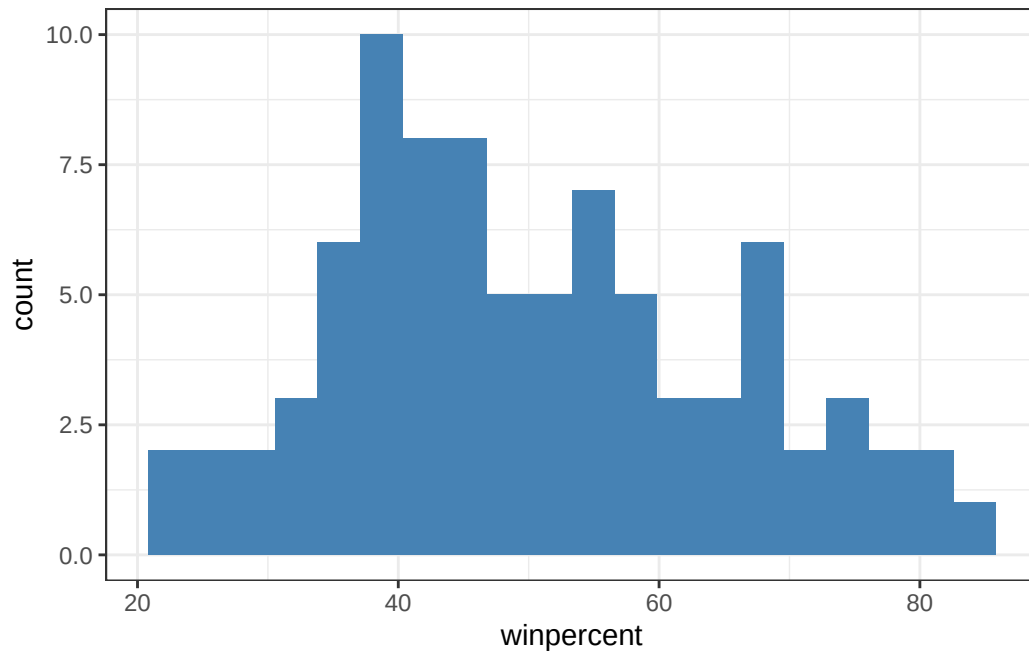
```
hist(candy$winpercent)
```

Histogram of candy\$winpercent



```
#ggplot version of histogram
library(ggplot2)

ggplot(candy, aes(winpercent)) +
  geom_histogram(bins=20, fill="steelblue") +
  theme_bw()
```



Q9. Is the distribution of winpercent values symmetrical?

No, it seems to be centered on lower percentages, with a tail towards higher percentages.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
median(candy$winpercent)
```

```
[1] 47.82975
```

If using the mean, then it is above 50%, however with a median use case it is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy[as.logical(candy$chocolate),]$winpercent)
```

```
[1] 60.92153
```

```
mean(candy[as.logical(candy$fruity),]$winpercent)
```

```
[1] 44.11974
```

On average chocolate candy is higher ranked than fruit candy.

Q12. Is this difference statistically significant?

```
t.test(candy[as.logical(candy$chocolate),]$winpercent, candy[as.logical(candy$fruity),]$winpercent)
```

Welch Two Sample t-test

```
data: candy[as.logical(candy$chocolate),]$winpercent and candy[as.logical(candy$fruity),]$winpercent
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Yes they seem to be different, as the p-value is much much lower than 0.05.

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
library("dplyr")
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
head(candy %>%
  arrange(winpercent), 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1		0.197	0.976
Boston Baked Beans				0	0	0	1		0.313	0.511
Chiclets				0	0	0	1		0.046	0.325
Super Bubble				0	0	0	0		0.162	0.116
Jawbusters				0	1	0	1		0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy %>%
  arrange(desc(winpercent)), 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup				0	0	0		0.720

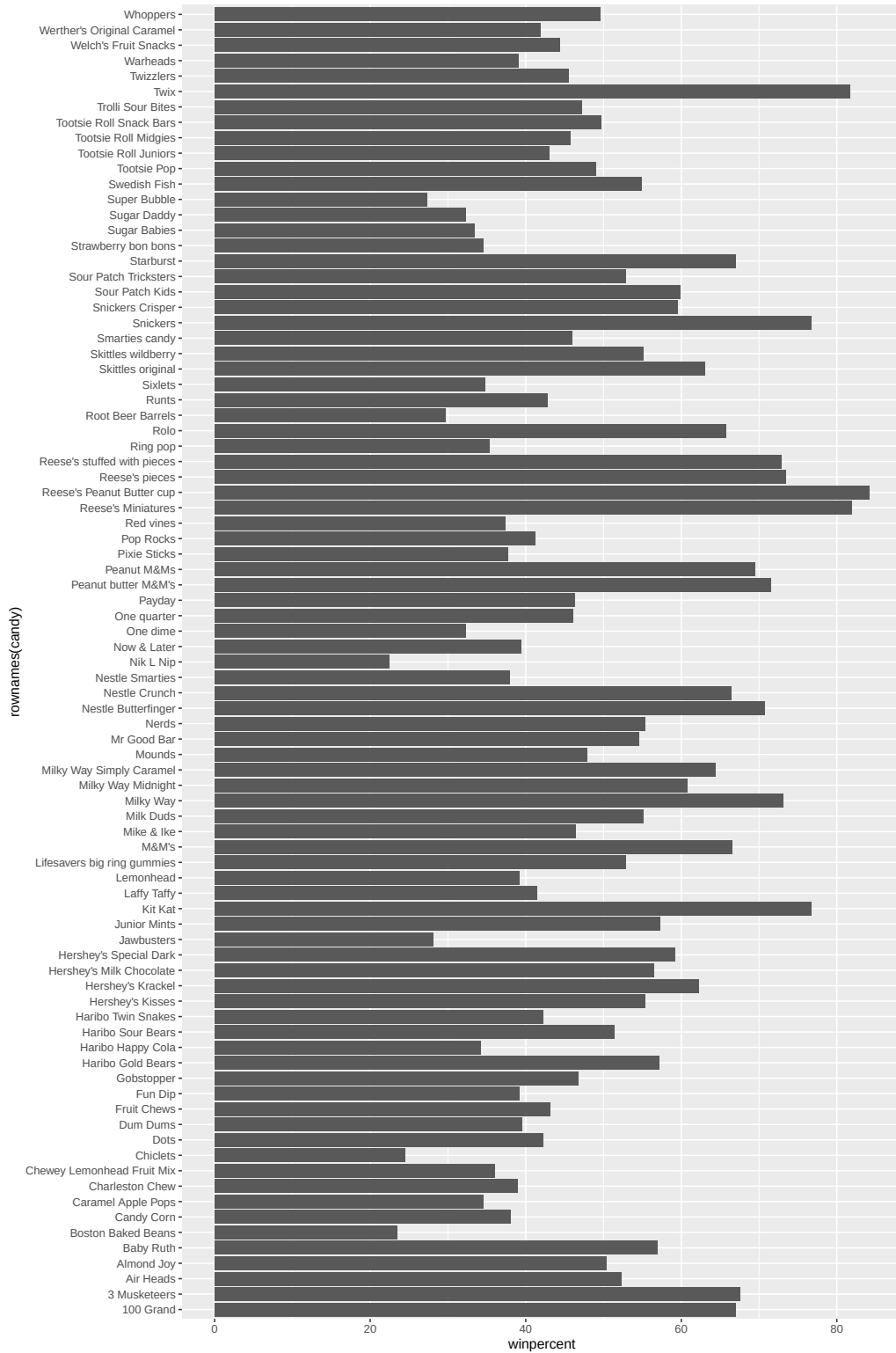
Reese's Miniatures	0	0	0	0	0.034
Twix	1	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Snickers	0	0	1	0	0.546

	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, Snickers

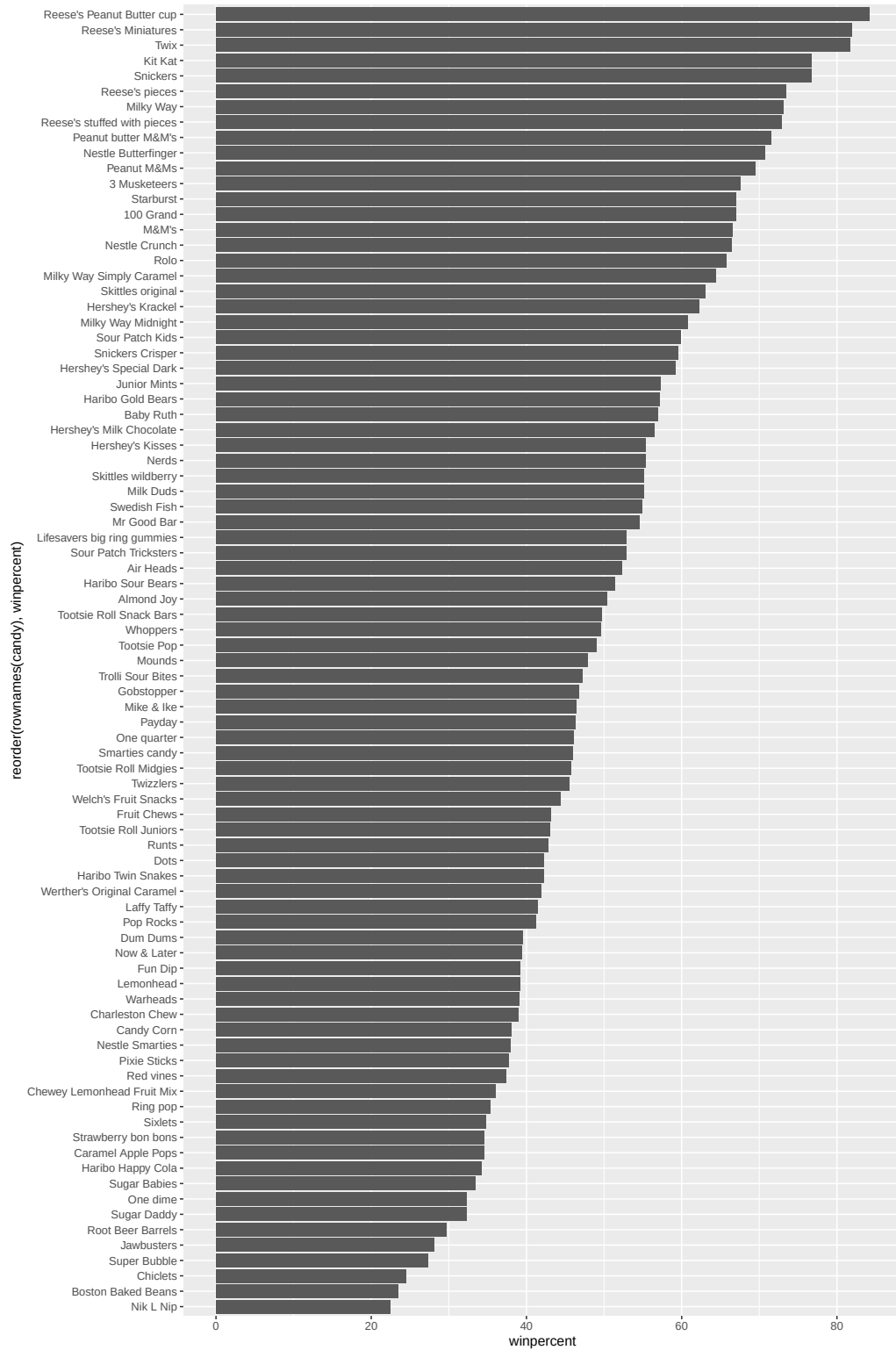
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy, aes(winpercent, rownames(candy))) +  
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

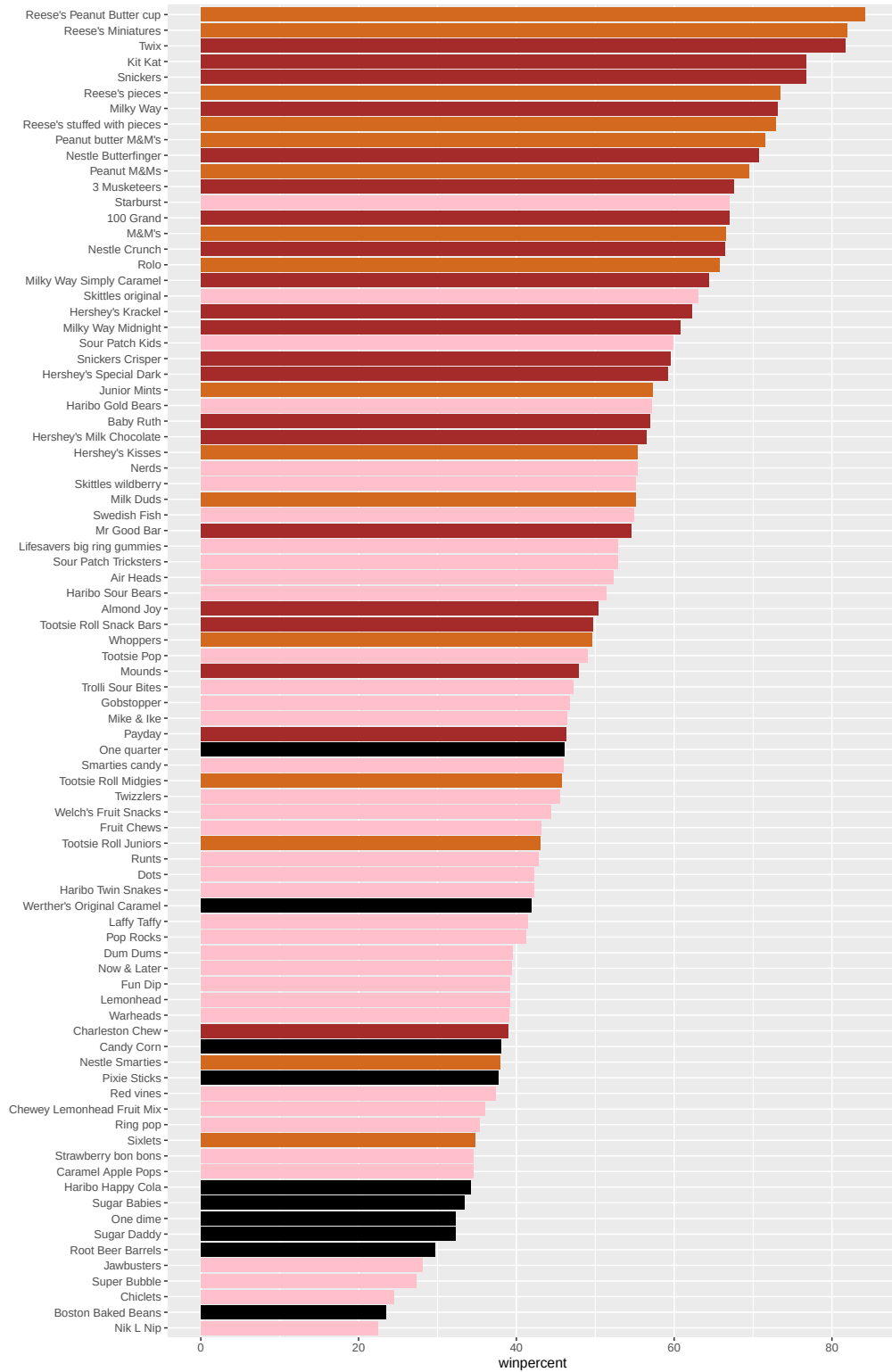
```
ggplot(candy, aes(winpercent, reorder(rownames(candy), winpercent))) +  
  geom_col()
```



Below I'll be able to change the colors so that it lines up with the type.

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$fruity)] = "pink"
my_cols[as.logical(candy$bar)] = "brown"
```

```
ggplot(candy, aes(winpercent, reorder(rownames(candy), winpercent))) +
  geom_col(fill=my_cols) +
  ylab("")
```



Q17. What is the worst ranked chocolate candy?

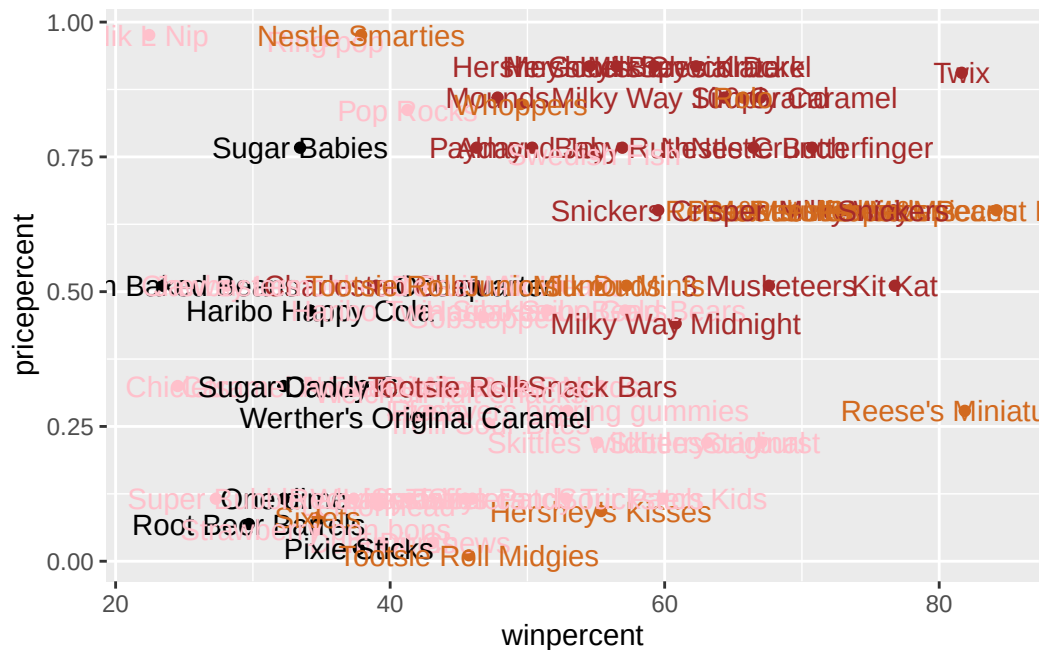
Sixlets

Q18. What is the best ranked fruity candy?

Starburst

Taking a look at pricepercent

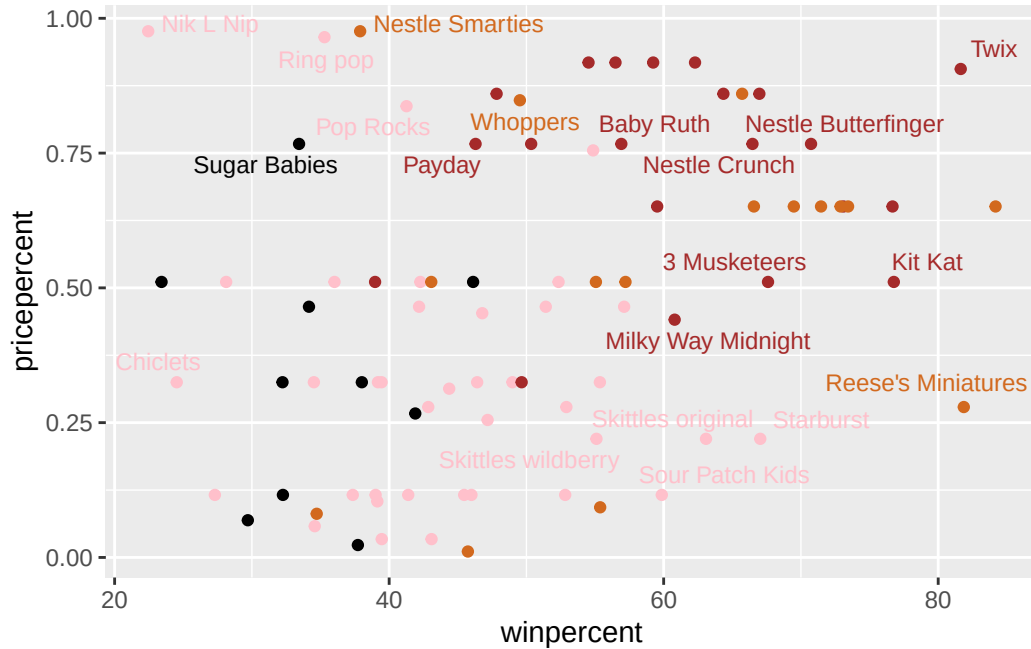
```
# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text(col=my_cols)
```



We can fix the label text overplotting with an add-on package called **ggrepel** and its `geom_text_repel()`

```
# install.packages("ggrepel")
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures seem to be the highest ranked for the least amount of money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
head(candy %>%
  arrange(desc(candy$pricepercent)))
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0	0	0	0

Nestle Smarties	1	0	0	0	0
Ring pop	0	1	0	0	0
Hershey's Krackel	1	0	0	0	0
Hershey's Milk Chocolate	1	0	0	0	0
Hershey's Special Dark	1	0	0	0	0
	crispedricewafer	hard	bar	pluribus	sugarpercent
Nik L Nip	0	0	0	1	0.197
Nestle Smarties	0	0	0	1	0.267
Ring pop	0	1	0	0	0.732
Hershey's Krackel	1	0	1	0	0.430
Hershey's Milk Chocolate	0	0	1	0	0.430
Hershey's Special Dark	0	0	1	0	0.430
	pricepercent	winpercent			
Nik L Nip	0.976	22.44534			
Nestle Smarties	0.976	37.88719			
Ring pop	0.965	35.29076			
Hershey's Krackel	0.918	62.28448			
Hershey's Milk Chocolate	0.918	56.49050			
Hershey's Special Dark	0.918	59.23612			

Nik L Nip, Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate are most expensive. The least popular of these by far is Nik L Nip.

Exploring the correlation structure

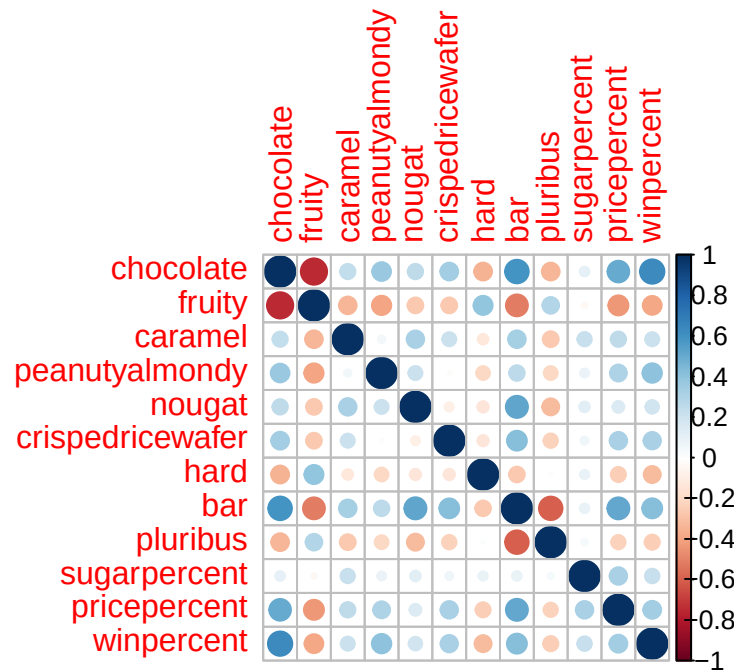
We can calculate the pair-wise correlation of all our columns

```
cij <- cor(candy)

# install.packages("corrplot")
library(corrplot)
```

corrplot 0.95 loaded

```
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Fruity and Chocolate seem to be anti-correlated, and fruity and bar seem to have a strong negative correlation as well.

Q23. Use your corplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

Likely fruity vs chocolate, bar vs pluribus, and winpercent.

Principal Component Analysis

In this case we want to be sure to set `scale=TRUE` because we have one variable `winpercent` that is on a very different scale than all others and would otherwise dominate our PCA results.

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

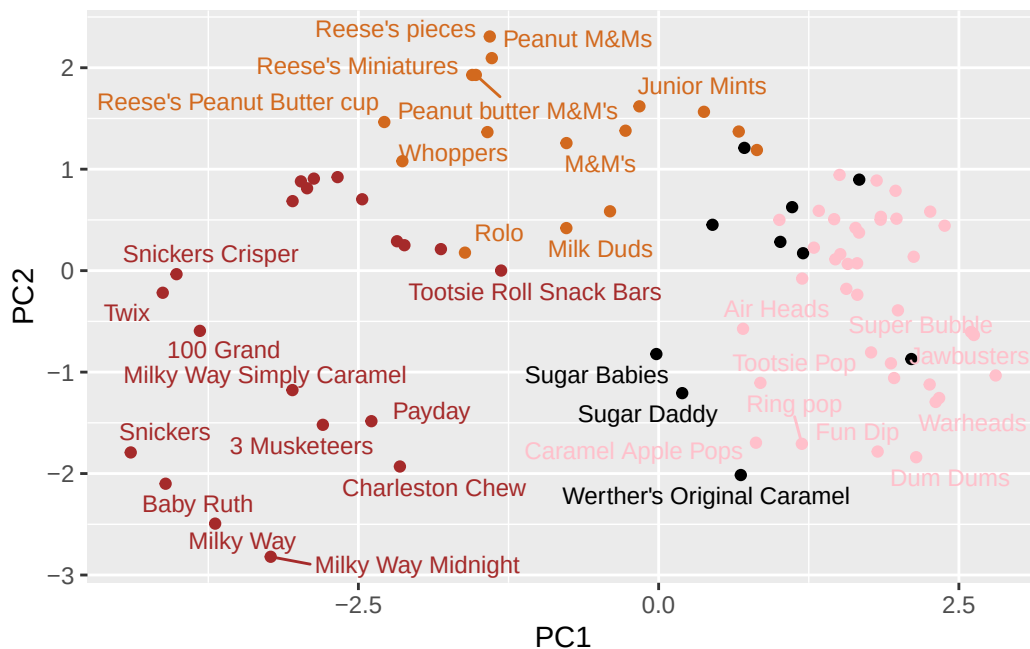
Importance of components:

PC1 PC2 PC3 PC4 PC5 PC6 PC7

Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369
	PC8	PC9	PC10	PC11	PC12		
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760		
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317		
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000		

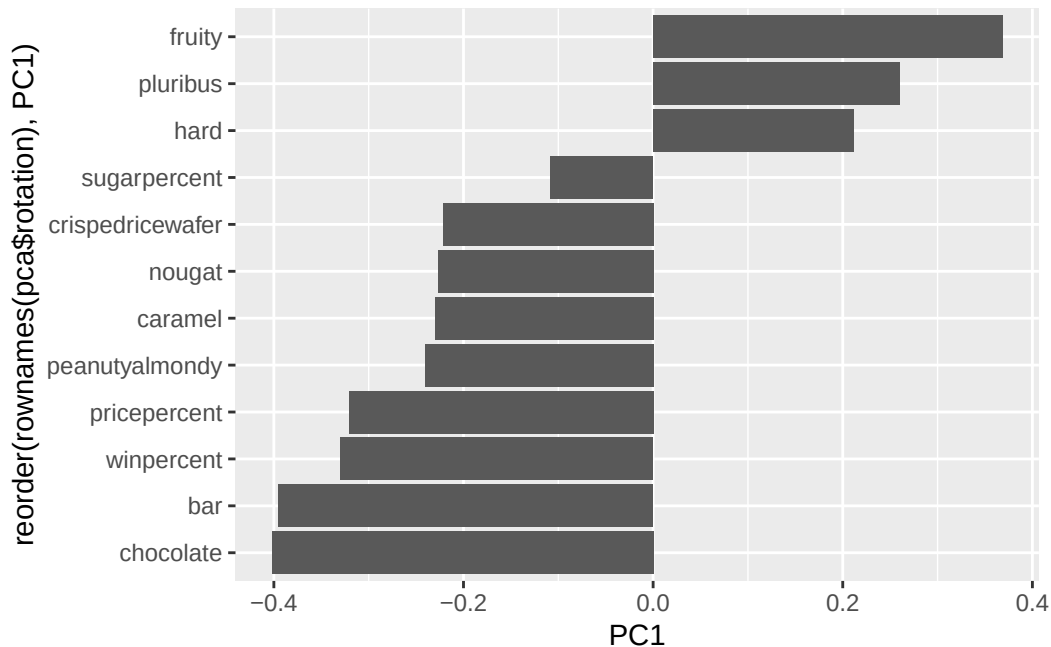
First major result figure is the “score plot” of PC1 vs PC2

```
ggplot(pca$x, aes(PC1, PC2, label=row.names(pca$x))) +
  geom_point(col=my_cols) +
  geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7)
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

```
ggplot(pca$rotation) +
  aes(PC1, reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



It seems that fruity, pluribus, and hard are picked up by PC1 in the positive direction very strongly, which makes sense as they seem to be correlated together on the chart, with many fruit candies like skittles being pluribus. This seems to have been shown well on the corrplot as well. It then places some pluribus candies like M&Ms and kisses in the middle, while having bar candies on the far left.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

It seems that chocolate based candies, especially bars, that contain peanut and almond seem to be the combination for a winning candy according to the corrplot. All of these traits positively correlate with winpercent thus they are a useful combination. The histogram shows the same trend, with chocolate candies and bar candies trending much higher on average than the fruit candies.